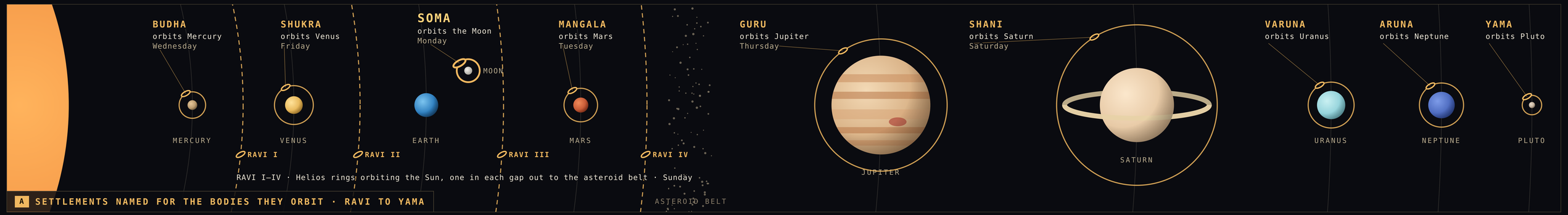
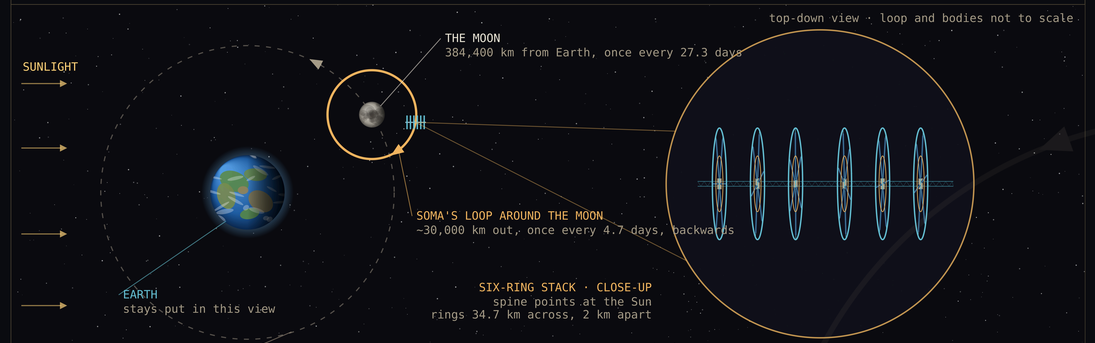
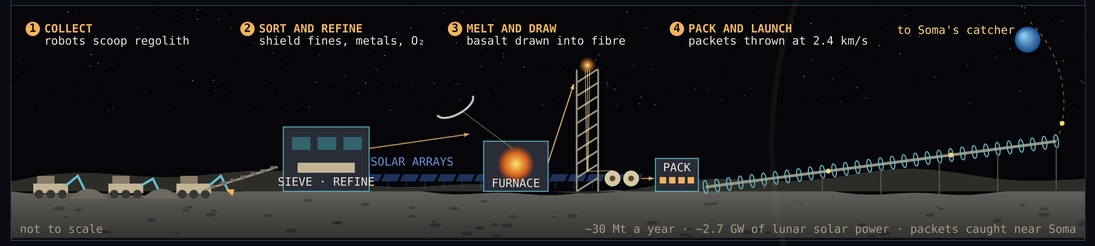
A SETTLEMENTS NAMED FOR THE BODIES THEY ORBIT · RAVI TO YAMA



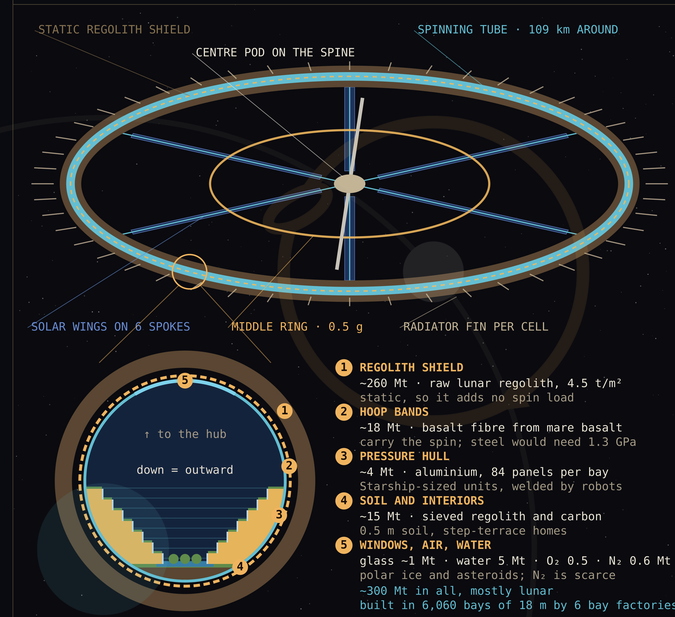
B SOMA, FINISHED: SIX RINGS ON ONE SPINE, LOOPING THE MOON



C ON THE MOON: REGOLITH TO BASALT FIBRE TO THE MASS DRIVER



D ONE RING, AND HOW IT'S MADE



1. **REGOLITH SHIELD** — ~260 Mt · raw lunar regolith, 4.5 t/m² static, so it adds no spin load
2. **HOOP BANDS** — ~18 Mt · basalt fibre from mare basalt carry the spin; steel would need 1.3 GPa
3. **PRESSURE HULL** — ~4 Mt · aluminium, 84 panels per bay Starship-sized units, welded by robots
4. **SOIL AND INTERIORS** — ~15 Mt · sieved regolith and carbon 0.5 m soil, step-terrace homes
5. **WINDOWS, AIR, WATER** — glass ~1 Mt · water 5 Mt · O_2 0.5 · N_2 0.6 Mt polar ice and asteroids; N_2 is scarce ~300 Mt in all, mostly lunar built in 6,060 bays of 18 m by 6 bay factories

E INSIDE: 45 LEVELS, 11 KINDS OF CELL

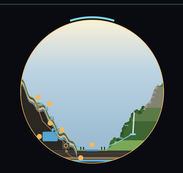


- **HOME** — terraced homes
- **LAB** — labs round a light well
- **FARM** — 9 growing tiers
- **WILD** — forest hills, river
- **WATER** — lake and beaches
- **FOOD** — food halls, night
- **PLAY** — fields, stands, wall
- **CIVIC** — schools, hospital
- **PLANT** — air, water, power
- **MAKE** — fab and repair
- **HUB** — spoke elevators

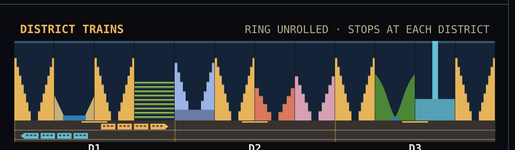
45 LEVELS — step terraces wherever people live

606 cells of 180 m in 30 districts; ~550,000 people per ring, 30% kept for parks, water and wild land.

F LAND, WATER AND TRANSIT

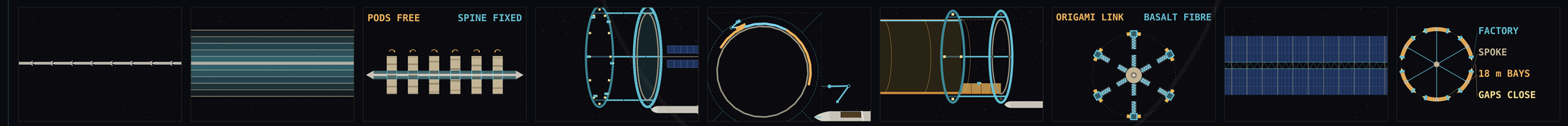


Hollow mountains filter the water down to a clean tank.



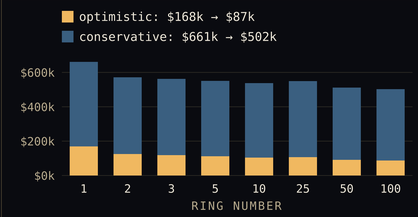
District trains run in the hardware base and link all 30 districts. They stay slow (~70 km/h): riding with the spin makes you heavier.

G TORUS BUILD



1. **The spine** — 280 Starships dock nose to tail; the end ships steer.
2. **Thicken** — Drums over the Starship core carry the load once rings spin.
3. **Centre pods** — Six pods, one per ring, turn freely on magnetic bearings over the fixed spine until their ring is attached.
4. **Factory setup** — Two rail rings, six beams, grippers and weld crawlers: 7 flights.
5. **Hanging the panels** — A loading arm unloads panels from the Starship; carts run them around the factory jig to hang.
6. **Building the tube** — The factory grips, welds a bay, then inchworms 18 m forward.
7. **Spokes** — Basalt-fibre origami links unfold from the spine pod and the tube's connector cell, and lock mid-span.
8. **Wings and fins** — 17 km² of solar wings; one radiator fin per cell.
9. **Half-built torus** — Factories at the six spokes lay tube both ways until the 12 fronts meet.

H COST PER RESIDENT



Cost per person falls as factories are reused and Earth parts get cheaper.

SOMA

A CISLUNAR RING SETTLEMENT

550,000	3.3 M	1 g	30,000 km	$87–661k
PEOPLE PER RING	PER SIX-RING STACK	AT 0.23 RPM	LOOP AROUND THE MOON	BUILD COST PER RESIDENT

ORDER-OF-MAGNITUDE ESTIMATES
SPECULATIVE DESIGN STUDY · 2026
ILLUSTRATIONS NOT TO SCALE